

Mesozoic radiolarians from the Khabarovsk complex, eastern USSR : their significance in relation to the Mino terrane, central Japan

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Key words : paleontology, radiolarians, Triassic, Jurassic, Khabarovsk, terrane

Introduction In the course of the geotectonic studies on "geosynclines" in Northeast Asia, Japanese geologists (*e.g.*, Kobayashi, 1984 ; Noda, 1956) have focused their interests on the complicated sedimentary formations in the Sikhote-Alin region in relation to the Japanese "Paleozoic" stratigraphy. Especially, Noda (1956) introduced the occurrence of Radiolaria-bearing siliceous formations near Khabarovsk, although he regarded the rocks as the Permian. After that, Russian paleontologists and biostratigraphers have reported Mesozoic radiolarians from the Sikhote-Alin Range, including the Khabarovsk area (*e.g.*, Zhamoida, 1972 ; Tikhomirova, 1986) ; most of their works, however, were based on observations under the optical microscope.

In this article, we describe the occurrence of Triassic and Jurassic radiolarians from the Khabarovsk complex, eastern USSR (stipple pattern in the inset map of Fig. 1), and discuss paleontological, stratigraphical and structural similarities between the Khabarovsk complex and the Mino terrane in central Japan.

Geologic setting and radiolarian fossils
The Khabarovsk complex has characteristics of

melanges consisting of blocks of basic volcanic rocks, limestone, chert, shale and sandstone in shaly matrix, and it can be seen along the Amur River bank in the Khabarovsk area (Fig. 1), where from north to south melange (I), chert (I), red shale (I), gray shale, chert (II), red shale (II), melange (II) and schist are exposed ; each rock unit is bounded

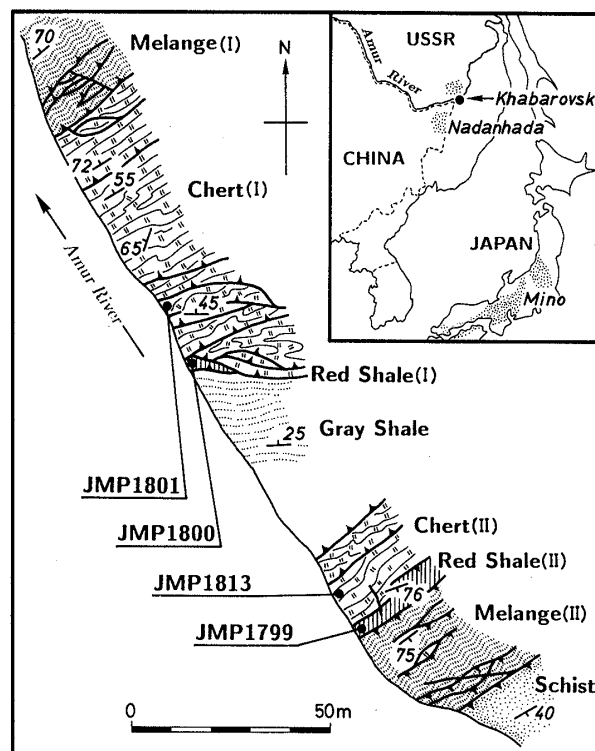


Fig. 1. Geologic sketch map of the Khabarovsk complex. Thick lines with sawteeth represent faults and their dip, and numbers (JMP1799, etc.) indicate localities of Radiolaria-bearing samples. In the inset map, the distribution of the Khabarovsk complex, and the Nadanhada and Mino terranes is roughly shown by stipple pattern. Modified from Natal'in and Zyabrev (1989).

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by faults. Melanges in this outcrop are composed of blocks of pillow basalt, limestone, chert and sandstone embedded in a sheared black shale matrix. The limestone blocks are believed to be late Carboniferous or Permian in age on the basis of fusulinids found in other localities (Natal'in & Zyabrev, 1989). Cherts, reddish brown in color, are characterized by rhythmic interbeddings of chert and shale. Shales are massive, homogeneous, and grayish red and dark gray to greenish gray in color. Middle(?) Triassic radiolarians such as *Triassocampe* spp. and *Pseudostylosphaera* spp. occur in the chert samples (JMP1801, 1813 in Fig. 1), whereas samples, JMP1799 (red shale II) and JMP1800 (red shale I), yield Jurassic radiolarians (Plate I). Radiolarian assemblage in JMP1799 contains *Hsuum hisuikyoense*, *H.*(?) sp. cf. *H.*(?) *matsuokai*, and *Eucyrtidiellum* sp. cf. *E. disparile*, and is assignable to the *Lactorum*(?) *jurassicum* Zone (upper Lower Jurassic) defined by Matsuoka & Yao (1986). This assignment is also supported by morphological changes of *Eucyrtidiellum* studied by Nagai & Mizutani (1990). Red shale sample JMP1800 bears *Eucyrtidiellum unumaense*, *Stichocapsa japonica*, *Tricolocapsa* sp. cf. *T. plicarum*, *T.*(?) *fusiformis*, *Trillus* sp., *Zartus* sp., etc; these species are characteristic of the radiolarian assemblage of *Tricolocapsa plicarum* Zone (lower Middle Jurassic) of Matsuoka & Yao (1986). The radiolarian fossils indicate that the original succession of the Khabarovsk complex in this area is chert, red shale (II) and red shale (I), in ascending order. The chert-shale units occur repeatedly along the fault just on the north of chert (II). Alternating beds of psammitic and pelitic schists lying in the southern margin of the mapped area (Fig. 1) consist of quartz, albite, muscovite and epidote, and the beds show well-developed foliation and lineation. Although the relationship between the schists and the above-mentioned nonmetamorphosed sedimentary rocks is unclear, the schists might be a rock unit originally unrelated to the Khabarovsk complex and juxtaposed during the later process of deformation.

Discussion The radiolarian assemblage from the Khabarovsk complex described above is very similar to those from the Mino terrane in central

Japan; most of the radiolarian genus and species are common to the Mino terrane (*e.g.*, Yao, 1979; Mizutani & Koike, 1982; Isozaki & Matsuda, 1985; Matsuoka & Yao, 1986; Nagai & Mizutani, 1990; unpublished data in the radiolarian data base: the Rad File, Dept. Earth Sci., Nagoya Univ.). Lithologic associations, such as Carboniferous to Permian limestone, Triassic bedded chert, early to middle Jurassic radiolarian siliceous shale, melanges with blocks of pillow basalt, chert, limestone and sandstone are common in the Khabarovsk and Mino regions (*e.g.*, Wakita, 1988; Mizutani, 1990), although no schistose rocks have been reported from the Mino terrane. Moreover, the geologic structure, for example, repetitive occurrence of Triassic bedded chert and early to middle Jurassic siliceous shale of the Khabarovsk complex seen along the Amur River, is regarded as one of the distinctive characteristics of the Mino terrane (Kojima, 1989). All these above mentioned characteristics demonstrate that the Khabarovsk complex in northeastern USSR has close relations with the Mino terrane in central Japan. Same kind of affinities between the Mino terrane and the Nadanhada terrane in Northeast China were reported by Mizutani *et al.* (1986), Kojima & Mizutani (1987), Kojima (1989) and Zhang *et al.* (1989). The similarities of these terranes strongly suggest that the rocks of the terranes were deposited in a continuous oceanic basin, and are common in sedimentary environments and deformation styles.

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* in Japanese with English abstract.

** in Chinese with English abstract.

*** in Russian.

<地 名> Mino美濃 Nadanhada那丹哈達

Plate I Triassic and Jurassic radiolarians from the Khabarovsk complex, eastern USSR. The fossil individuals are numbered as (12345/6789); the numerator represents a sequential number of SEM photograph and the denominator indicates that of the rock specimen (JMP numbers shown in Fig. 1) from which the fossil was extracted. Both numbers are registered in the Rad File, Department of Earth Sciences, Nagoya University. 1; *Triassocampe* sp. cf. *T. deweveri*, (52654/1801). 2; *Pseudostylosphaera* sp., (53167/1813). 3; *Archaeodictyomitra* sp., (53174/1800). 4; *Wrangellium* sp. A, (53232/1800). 5; *Hsuum* sp. A, (53206/1800). 6; *Hsuum hisuikyoense*, (52592/1799). 7; *Ristola* sp. A, (53176/1800). 8; *Archicapsa pachyderma*, (53180/1800). 9; *Tricolocapsa*(?) sp. A, (53172/1800). 10; *Tricolocapsa* sp. cf. *T. plicarum*, (52573/1799). 11; *Tricolocapsa* sp. cf. *T. ruesti*, (53236/1800). 12; *Archaeodictyomitra*(?) sp. A, (53222/1800). 13; *Eucyrtidiellum unumaense*, (53229/1800). 14; *Tricolocapsa* sp. cf. *T. plicarum*, (52653/1800). 15; *Stichocapsa japonica*, (53173/1800). 16; *Stichocapsa*(?) sp. A, (53189/1800). 17; *Eucyrtidiellum* sp. cf. *E. disparile*, (52567/1799). Scale bar : 100 μ m for 8, 9, 10, 11, 13, 14, 15, 17; 150 μ m for 1, 3, 4, 5, 7, 12, 16; 205 μ m for 2, 6.

